

ABHISHEK THAKUR

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Summary

Software Engineer with **3.5+ years** building **.NET** backend systems, distributed event pipelines, data-intensive applications, and **Angular** web platforms. At I2V Systems, worked on a video analytics platform deployed across up to 10,000 cameras, improving event throughput, PostgreSQL/TimescaleDB performance, analytics exports, and vector search for face recognition and ANPR. At Eurofins, extending the same engineering foundation into **Elasticsearch**, **RAG**, **semantic search**, **embeddings**, and **LLM agent integrations** for an enterprise procurement platform.

Experience

Eurofins IT Delivery Center

October 2025 – Present

Software Engineer

- Developed **.NET** services for catalog search on **Elasticsearch**, combining exact SKU matching, trigram typo tolerance, scientific-term synonyms, and dense-vector similarity with Reciprocal Rank Fusion; synchronized search indexes with transactional writes through Change Data Capture.
- Extended the **RAG** pipeline for assay documentation using metadata-driven chunking and Azure OpenAI embeddings; moved document ingestion to **Hangfire** background jobs to isolate heavy processing from live traffic and added **Redis semantic caching** for repeat queries.
- Implemented a CI relevance evaluation pipeline using a golden-query set with **MRR** and **NDCG**, failing builds below 0.85 to catch search and RAG ranking regressions before release.
- Implemented **Model Context Protocol (MCP)** endpoints in ASP.NET Core, exposing procurement capabilities to internal LLM agents through typed tool schemas and role-based access control.
- Enhanced an existing headless **Umbraco** application with **Angular SSR** for SEO-critical pages, Redis-cached HTML, and AWS S3/CloudFront static asset delivery to improve SEO and page performance.

I2V Systems

April 2023 – September 2025

Software Engineer

- Contributed to an event-driven platform using **RabbitMQ**, **MassTransit**, and **Hangfire**, helping increase processing throughput from 20–30 to **500+ events/sec** across deployments supporting up to 10,000 cameras.
- Refactored a 40+ project **.NET 8** monolith toward Clean Architecture, reducing dependency violations by **90%** and production issues by **20%**.
- Built a **face recognition pipeline** using pgvector and Milvus with HNSW and IVF_SQ8 indexes; reduced ANN index memory by **75%** through scalar quantization, with face embeddings served through NVIDIA Triton.
- Developed ANPR pipelines and dashboards for **traffic activities, violations, and entry/exit monitoring**, including a Central Monitoring Hub aggregating events from **100+ servers**; optimized dashboard performance with **materialized views, scheduled preprocessing, and query tuning**, to enable fast, precise time-based event retrieval.
- Reworked a 100K+ record analytics export in **.NET and PostgreSQL**, **reducing export time by 92%** from 12 minutes to under 1 minute and enabling reliable execution on Linux.
- Improved a **TimescaleDB** ingestion pipeline for 400 cameras from 70 to **300+ frames/sec** through indexing and database tuning; diagnosed bottlenecks using PostgreSQL query plans and optimized inefficient EF Core queries.

Projects

Jarvis: AI Resume Tailoring Platform | Python, FastAPI, PostgreSQL/pgvector, Docker

- Built an LLM pipeline that rewrites LaTeX resumes for a target job description, using **Pydantic-validated structured outputs** at each stage and **pgvector cosine similarity** to rank job postings.

Technical Skills

Languages: C#, TypeScript, Python, SQL, JavaScript, Bash

Backend: ASP.NET Core, EF Core, .NET 8, SignalR, RabbitMQ, MassTransit, Hangfire

Databases & Search: PostgreSQL, TimescaleDB, Redis, Elasticsearch, MS SQL Server; query plan analysis, indexing, performance tuning

AI & Vector Search: RAG, pgvector, Milvus (HNSW, IVF_SQ8), Model Context Protocol, Azure OpenAI embeddings, NVIDIA Triton, semantic search, MRR/NDCG evaluation

Frontend: Angular (RxJS, Signals, SSR), Umbraco, TypeScript libraries

Cloud & Ops: Docker, Azure DevOps CI/CD, AWS (S3, CloudFront), Linux, Grafana, Git

Architecture & Testing: Clean Architecture, Domain-Driven Design, unit testing (MSTest, Karma/Jasmine)

Education

Maharaja Agrasen Institute of Technology

Sept 2019 – Aug 2023

B.Tech, Information Technology